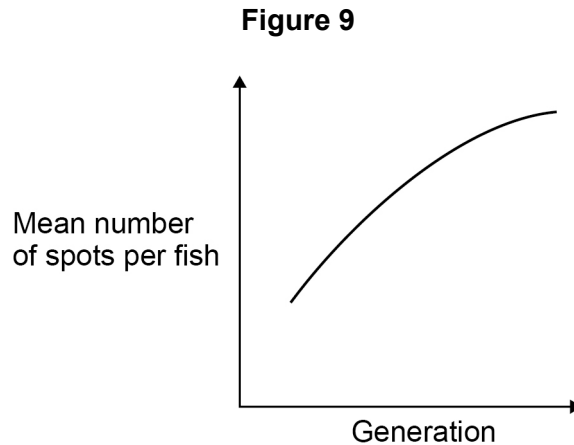


0 8

Guppies are small fish with brightly coloured spots on their body. The number of spots on a guppy is controlled by genes.

A scientist investigated guppy breeding in a large population of guppies in a fish tank. The fish tank contained brightly coloured stones. After each generation of breeding, she counted the number of spots on every guppy.

Figure 9 shows her results.



0 8 . 1

The scientist concluded that the mean number of spots on the guppies changed in this fish tank because the brightly coloured stones had affected the behaviour of the guppies.

The guppies did **not** behave aggressively towards each other and their feeding behaviour did **not** change.

Suggest **one** type of guppy behaviour that could be affected by the presence of brightly coloured stones.

[1 mark]



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Suggest:

- a further investigation the scientist could do to confirm that the brightly coloured stones had affected guppy behaviour
- a null hypothesis for the new investigation.

[2 marks]

Further investigation _____

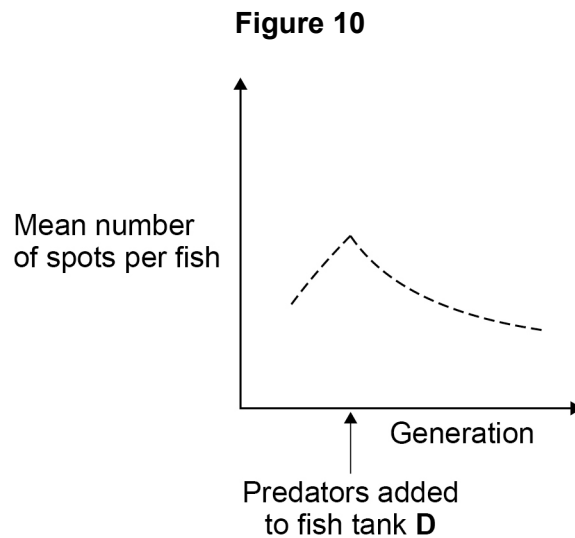
Null hypothesis _____

Question 8 continues on the next page**Turn over ►**

0 8 . 3

The scientist repeated the investigation of guppy breeding in an identical fish tank, but added predators of the guppies into the fish tank at the fourth generation of breeding.

Figure 10 shows her results.



Name the type of selection the scientist investigated in this fish tank.

Explain why this selection affected the frequency of alleles in the population of guppies in this fish tank.

[4 marks]

Type of selection _____

Explanation _____

